

Red Pitaya DDS IP Core with AXI-DMA Interface: Design, Verification, and Hardware Validation Report

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Abstract—This report presents the design, RTL simulation, and hardware validation of a dual-channel Direct Digital Synthesis (DDS) IP core targeting the Xilinx Zynq-7020 SoC on the Red Pitaya STEMLab 125-14 platform. The core features 48-bit phase accumulators, four built-in waveform types, arbitrary waveform playback via AXI-DMA with glitch-free dual-bank LUT switching, and a shadow/active register architecture for atomic multi-parameter updates. Icarus Verilog simulations verify the 7-stage pipeline, configuration FSM, and DMA sink FSM. Hardware measurements using a Siglent oscilloscope confirm sine SFDR of 65 dBc–70 dBc, frequency accuracy within 20 ppm, amplitude linearity $R^2 = 0.999995$, and dual-channel phase synchronisation error below 0.008° across four phase offsets.

Index Terms—DDS, FPGA, IP Core, AXI-DMA, Red Pitaya, Zynq-7020, Waveform Generator

I. INTRODUCTION

Direct Digital Synthesis (DDS) is a widely used technique for generating programmable waveforms in FPGA-based instrumentation [4]. This report documents `red_pitaya_dds_axi_dma`, a custom IP core designed for the Red Pitaya STEMLab 125-14 platform [2] (Zynq-7020 SoC [1], 14-bit DAC at 125 MS/s).

The core provides:

- Dual independent DDS channels with 48-bit phase accumulators, yielding sub-micro-hertz frequency resolution ($\Delta f = f_{\text{clk}}/2^{48} \approx 4.44 \times 10^{-7}$ Hz).
- Five waveform modes: sine (ROM), square, triangle, sawtooth, and arbitrary (phase-addressed LUT).
- Arbitrary waveform loading via Xilinx AXI-DMA (MM2S, direct-register mode) with dual-bank LUT architecture for glitch-free updates.
- Shadow/active register scheme ensuring atomic parameter application across both channels.
- AXI4-Lite register interface compatible with the Red Pitaya system bus.

The remainder of this report is organised as follows: Section II describes the system architecture and SoC integration; Section III details the DDS channel pipeline; Section IV covers the AXI-DMA subsystem; Section VI presents RTL simulation results; Section VII reports hardware validation measurements; and Section IX summarises findings.

II. SYSTEM ARCHITECTURE

A. SoC Platform

The target is a Xilinx Zynq-7020 SoC integrating dual ARM Cortex-A9 cores (Processing System, PS) with Artix-7

programmable logic (PL). The PS is responsible for parameter configuration, mathematical expression evaluation, waveform sample staging in DDR, and AXI-DMA control. The PL implements the real-time DDS datapath, DAC interface, oscilloscope acquisition, and PID controller.

B. AXI Bus Topology

Three distinct AXI paths connect the PS and PL (Figure 1):

- 1) **Control registers** (AXI3 via GP0, converted to AXI4): The PS `M_AXI_GP0` port (AXI3) connects through an `axi4_slave` bridge and an 8-way `sys_bus_interconnect` to peripheral register files. The DDS core occupies slot 2 at base address `0x4020_0000` with a 20 bit address window.
- 2) **DMA control** (AXI4-Lite): A dedicated AXI-Lite port at `0x8040_0000` provides direct-register-mode control of the Xilinx AXI-DMA IP [3] (MM2S only).
- 3) **DMA data** (AXI4 via HP port): The AXI-DMA reads waveform samples from a reserved DDR buffer at physical address `0x1E00_0000` and streams them via AXI4-Stream to the DDS core's AXIS sink, crossing into the PL clock domain through an `axis_clock_converter`.

C. Clock Architecture

The PLL (`red_pitaya_pll`) derives all PL clocks from the external 125 MHz ADC reference crystal:

- **adc_clk** (125 MHz): master sample clock for all DDS logic, phase accumulators, LUT RAMs, and configuration FSM.
- **dac_clk_1x** (125 MHz): DAC output register clock.
- **dac_clk_2x** (250 MHz): DAC DDR write strobe via ODDR primitive.
- **dac_clk_2p** (250 MHz, -45° phase): DAC DDR clock output via ODDR primitive.
- **FCLK_CLK0** (125 MHz, PS-sourced): AXI DMA and system bus interface clock.

Clock domain crossings are handled by the `sys_bus_interconnect` (`FCLK`→`adc_clk`) and an `axis_clock_converter` (`FCLK`→`adc_clk` for DMA stream data).

III. DDS CHANNEL PIPELINE

Each DDS channel (`rp_dds_channel`) implements a 7-stage synchronous pipeline clocked at 125 MHz. Key architectural parameters are summarised in Table I.

Figure 10 illustrates the dataflow of the 7-cycle pipeline. The stages and their outputs are as follows:

- S0 (Phase Acc):** 48b output.
- S1 (Phase Offset Add):** 48b output, 14b ctrl output.
- S2 (Waveform MUX):** 14b output.
- S3 (DSP48 Multiply):** 32b output.
- S4 (Arith. Shift):** 18b output.
- S5 (DC Offset Add):** 19b output.
- S6 (Saturate Output):** 14b output.

Annotations indicate data propagation:

- amplitude (16b) propagates S0 → S2, consumed at S3
- dc_offset (14b) propagates S0 → S6
- valid token propagates S0 → S6

All stages are clocked on `adc_clk` (125 MHz). Total latency: 7 cycles = 56 ns.

- *Sine* (0): synchronous ROM lookup from `rp_sine_rom`

TABLE I: DDS Channel Pipeline Parameters

Parameter	Value	Unit
Phase accumulator width	48	bits
Frequency tuning word (FTW)	48	bits
Frequency resolution	4.44×10^{-7}	Hz
Maximum output frequency	62.5	MHz
Sine ROM depth $\times$ width	4096×14	entries
Arb LUT depth $\times$ width	16384×14	entries
Amplitude word (Q1.15)	16	bits
DC offset	14	bits (signed)
DAC output	14	bits (signed)
Pipeline stages	7	cycles
Pipeline latency	56	ns
DSP48E1 usage	1	per channel

- *Square* (1): $\phi_{\text{eff}}[47] ? -8192 : +8191$
- *Triangle* (2): folded-ramp function of the upper phase bits
- *Sawtooth* (3): $\phi_{\text{eff}}[47:34]$ reinterpreted as signed 14-bit
- *Arbitrary* (4): phase-addressed LUT bank readback

S3 — DSP Multiply. Sign-extends the sample to 15 bits and the amplitude to 17 bits, then computes the full 32-bit product. A (* use_dsp = "yes" *) attribute ensures DSP48E1 inference.

S4 — Scale. Arithmetic right-shift by 15 converts the Q1.15 amplitude product back to the 14-bit sample range (18-bit intermediate).

S5 — DC Offset. Adds the 14-bit signed DC offset to the 18-bit scaled sample (19-bit sum).

S6 — Saturation and Output. Clamps the 19-bit sum to the $[-8192, +8191]$ signed 14-bit range and registers the final DAC output word.

B. Shadow/Active Register Architecture

All per-channel parameters (enable, waveform select, FTW, phase offset, amplitude, DC offset, active LUT bank) are double-buffered. The PS writes *shadow* registers at any time without affecting the live output. A single write to `REG_CONTROL[0]` triggers the configuration FSM, which copies all shadow values to the active set in one clock cycle. This guarantees atomic multi-parameter updates across both channels.

IV. AXI-DMA SUBSYSTEM

A. Arbitrary Waveform Loading

Arbitrary waveform playback uses a dual-bank LUT architecture with four independent BRAM banks (A0, A1, B0, B1), each storing 16384×14 -bit samples. The PS loads waveform data into the *inactive* bank via AXI-DMA while the *active* bank continues driving the output, enabling glitch-free waveform updates.

The loading sequence is:

- 1) PS evaluates the mathematical expression and samples 16384 points normalised to $[-1, 1]$, then quantises to signed 14-bit integers.
- 2) PS writes the sample buffer to reserved DDR at `0x1E00_0000`.
- 3) PS configures the DDS core: target channel, target (inactive) bank, expected word count (16384).

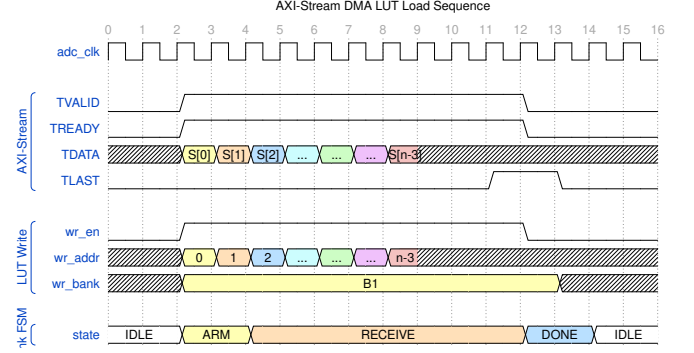


Fig. 3: AXI4-Stream DMA LUT load timing. The sink FSM transitions from IDLE through ARMED and RECEIVE to DONE as stream beats are accepted. Each beat writes one 14-bit sample to the target LUT bank.

- 4) PS arms the AXIS sink (`REG_DMA_CONTROL[0]`).
- 5) PS programs AXI-DMA MM2S: source address, byte count ($4 \times 16384 = 65\,536$ B).
- 6) AXI-DMA reads DDR and streams 32-bit words via `M_AXIS_MM2S`.
- 7) `axis_clock_converter` bridges to the `adc_clk` domain.
- 8) `rp_dds_dma_axis_sink` writes each sample (bits [13:0]) sequentially into the target LUT bank.
- 9) PS verifies completion, then sets the active bank to the newly loaded bank and triggers apply.

At 125 MHz with sustained streaming, the minimum bank fill time is $16384/125 \times 10^6 \approx 131 \mu\text{s}$.

B. AXIS Sink FSM

The `rp_dds_dma_axis_sink` implements a 5-state FSM (Figure 3): `IDLE` $\rightarrow$ `ARMED` $\rightarrow$ `RECEIVE` $\rightarrow$ `DONE` $\rightarrow$ `IDLE`, with an `ERROR` state for protocol violations. `TREADY` is only asserted in `ARMED` and `RECEIVE` states. The FSM verifies frame integrity by checking `TLAST` alignment against the expected word count.

C. Configuration FSM

The `rp_dds_cfg_fsm` sequences shadow-to-active register transfers and optional phase accumulator resets (Figure 4). For an apply-only operation, the sequence completes in 3 clock cycles (24 ns); for apply-with-phase-clear, 4 cycles (32 ns).

V. REGISTER MAP

Table II summarises the global and DMA registers. Per-channel registers (Table III) follow an identical layout at offsets `0x100` (channel A) and `0x200` (channel B). Figure 5 shows a representative AXI4-Lite register write transaction (apply command).

VI. RTL SIMULATION RESULTS

RTL simulations were performed using Icarus Verilog 12.0. Four testbench configurations were exercised, all passing without errors.

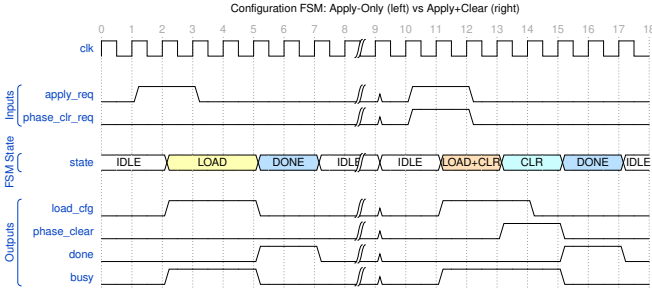


Fig. 4: Configuration FSM timing. Left: apply-only (3 cycles). Right: apply with phase clear (4 cycles). The `load_cfg` pulse triggers the atomic shadow→active register copy.

TABLE II: Global and DMA Register Map (Base: 0x4020_0000)

Offset	Name	R/W	Description
0x000	ID	R	Core ID (0x44445332)
0x004	VERSION	R	0x00010000
0x008	CONTROL	W	[0] apply, [1] phase_clr
0x00C	STATUS	R	10-bit status (Table IV)
0x010	SAMPLE_RATE	R	125 000 000
0x014	LUT_LENGTH	R	16 384
0x018	FEATURES	R	0x00000007
0x01C	DMA_CTRL	W	[0] arm, [1] clear, [2] abort
0x020	DMA_TARGET	W/R	[0] channel, [1] bank
0x024	DMA_EXPECT	W/R	Expected stream words
0x028	DMA_RCVD	R	Received word count
0x02C	DMA_ERR	R	Error code (0–5)

TABLE III: Per-Channel Register Map (Offsets from channel base)

Offset	Name	R/W	Description
+0x00	CTRL	W/R	[0] enable, [3:1] wave_sel
+0x04	FTW_LO	W/R	FTW bits [31:0]
+0x08	FTW_HI	W/R	FTW bits [47:32]
+0x0C	PHASE_LO	W/R	Phase offset [31:0]
+0x10	PHASE_HI	W/R	Phase offset [47:32]
+0x14	AMP	W/R	Signed 16-bit (Q1.15)
+0x18	DC	W/R	Signed 14-bit offset
+0x1C	ARB_BANK	W/R	Active LUT bank (bit 0)

TABLE IV: STATUS Register Bit-Field Definition (0x00C)

Bit	Name	Description
0	cfg_busy	Configuration FSM active
1	cfg_done	Configuration completed
2	dma_armed	AXIS sink armed
3	dma_busy	AXIS sink receiving
4	dma_done	DMA transfer completed
5	dma_error	DMA protocol error
6	cha_bank	CHA active LUT bank
7	chb_bank	CHB active LUT bank
8	cha_enable	CHA enabled
9	chb_enable	CHB enabled

A. Waveform Generation Verification

Figure 6 shows the simulated DAC output for all four built-in waveform types at $FTW = 2^{40}$ (corresponding to 488.28 kHz at 125 MHz). The simulation confirms correct waveshape generation for all modes with approximately 256 samples per period.

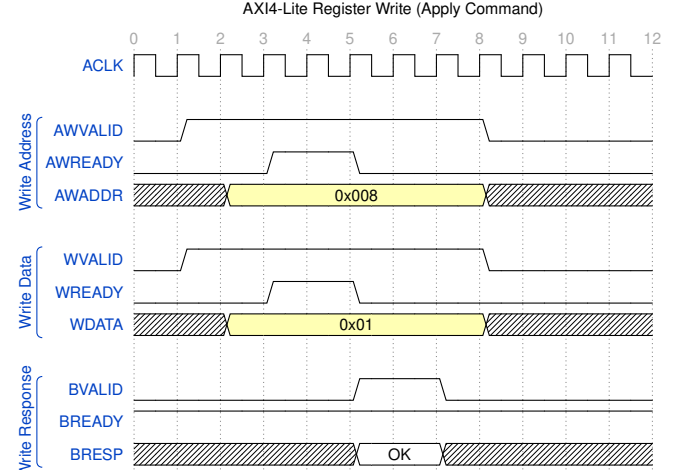


Fig. 5: AXI4-Lite register write transaction. The PS writes 0x01 to REG_CONTROL (0x008) to trigger the apply command.

DDS Channel Pipeline -- Simulated Waveform Outputs

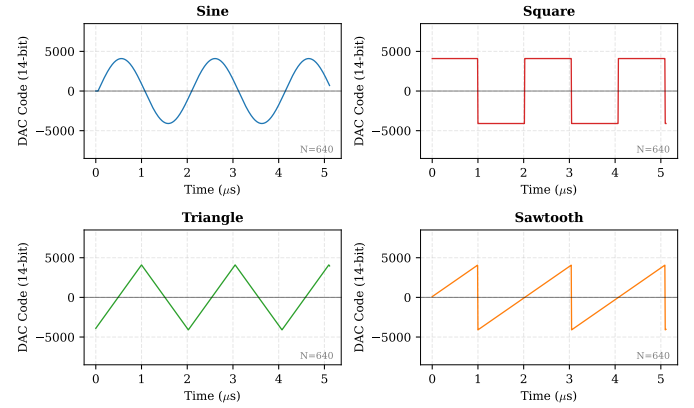


Fig. 6: Simulated DDS output waveforms for all four built-in modes. $FTW = 2^{40}$, amplitude = 16383 (near full-scale Q1.15). Each subplot shows ~ 2.5 periods.

B. Spectral Analysis

Figure 7 presents the FFT spectrum of the simulated sine output computed using a Blackman–Harris window. The measured SFDR is 87.6 dBc, with the fundamental at -9.0 dBFS (consistent with the half-scale amplitude setting). This confirms the 14-bit sine ROM provides excellent spectral purity in simulation.

C. FSM Timing Verification

Figure 8 validates the configuration FSM timing across four test sequences: apply-only, clear-only, combined apply+clear, and staggered operations. The `load_cfg` and `phase_clear` output pulses are correctly generated as single-cycle assertions in the expected states.

D. DMA Transfer Verification

Figure 9 shows a complete 4-word AXI-Stream DMA transfer through the AXIS sink FSM. The FSM correctly transitions

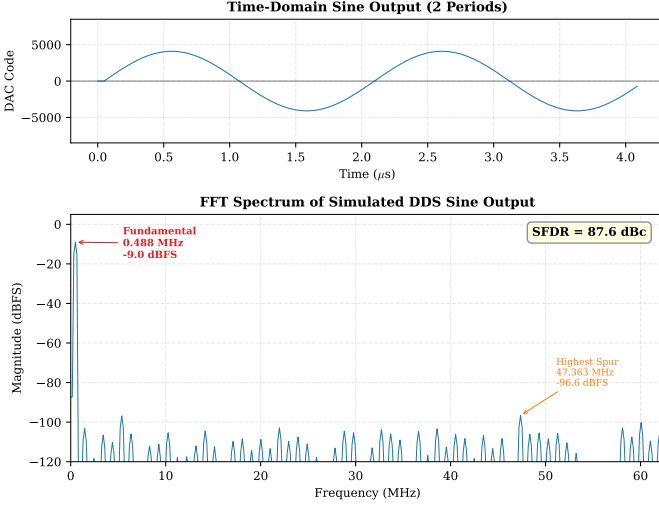


Fig. 7: Simulated sine FFT spectrum. SFDR = 87.6 dBc. The fundamental at 488 kHz dominates; the highest spur at 47.4 MHz is -96.6 dBFS, likely a windowing artefact.

Configuration FSM -- State Transition Timing Diagram

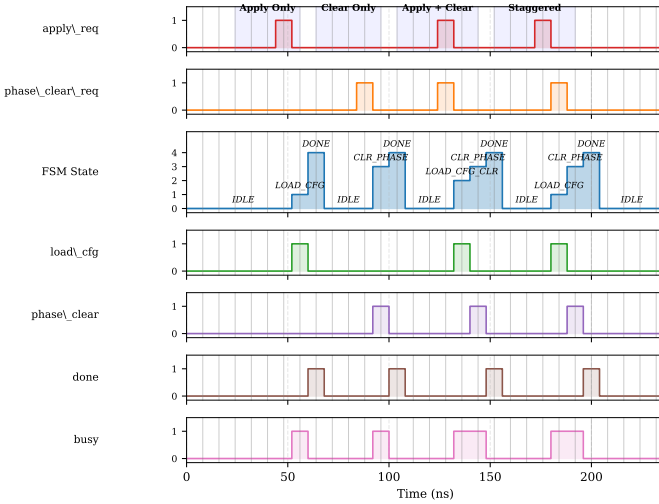


Fig. 8: Simulated configuration FSM state transitions. Four test sequences verify all state paths. One-cycle output pulses for load_cfg, phase_clear, and done are confirmed.

IDLE→ARMED→RECEIVE→DONE, with wr_en asserting on each accepted beat and TLAST alignment verified on the final word.

VII. HARDWARE VALIDATION RESULTS

Hardware measurements were performed on a Red Pitaya STEMLab 125-14 with DAC outputs captured by a Siglent oscilloscope. The test campaign comprised 34 test cases across four categories.

A. Test Setup

- **DUT:** Red Pitaya STEMLab 125-14 (Zynq-7020, XC7Z020)
- **Bitstream:** red_pitaya_dds_axi_dma on PL fabric

DMA AXI-Stream Sink -- 4-Word Transfer Sequence

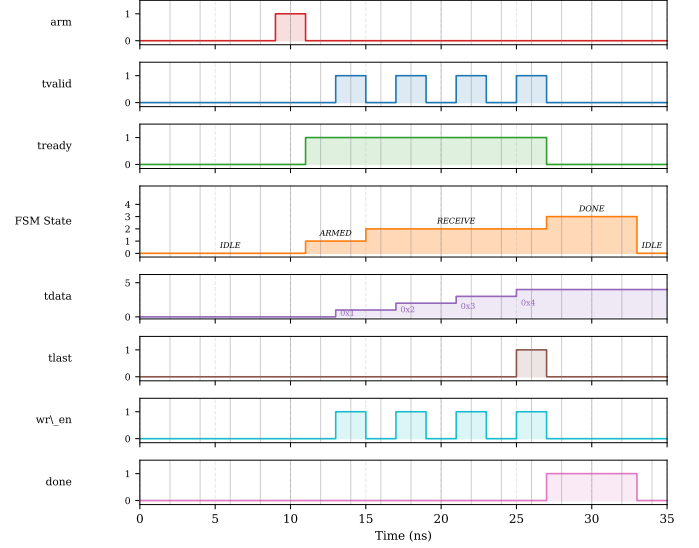


Fig. 9: Simulated AXI-Stream DMA transfer (4 words). TVALID/TREADY handshake, sequential wr_addr increment, and TLAST framing are verified.

- **Scope:** Siglent SDS (2 channels, direct BNC connection)
- **Control:** Python DDS client via HTTP API
- **Automated test runner:** 34 test cases in run_tests.py
- **Analysis:** Automated spectral analysis via analyze.py

B. Preset Waveform Measurements

Figure 10 presents oscilloscope captures of all 20 preset waveform configurations (4 types × 5 frequencies from 100 Hz to 1 MHz). All waveforms are cleanly reproduced across the full frequency range. At 1 MHz, bandwidth-limiting effects are visible on the sawtooth and square waveforms, as expected given the 125:1 oversampling ratio.

C. Spectral Purity

Figure 11 shows the measured total harmonic distortion (THD) as a function of frequency for all four waveform types. Sine THD remains below 0.07% across the entire frequency range, confirming the ROM-based lookup produces spectrally clean output. The non-sinusoidal waveforms exhibit THD consistent with their theoretical Fourier series content: ~12% for triangle, ~39% for square, and ~70% for sawtooth.

Figure 12 presents the FFT spectrum of the measured 100 kHz sine output computed using a Blackman–Harris window. Across all sine preset measurements, the SFDR ranges from 65.5 dBc to 69.6 dBc (Table VI). Compared to the simulation SFDR of 87.6 dBc (Figure 7), the hardware measurement includes DAC quantisation noise, board-level coupling, and the oscilloscope’s own noise floor contributions.

D. Amplitude Linearity

Figure 13 plots measured peak-to-peak voltage versus commanded amplitude for 100 kHz sine at four amplitude settings

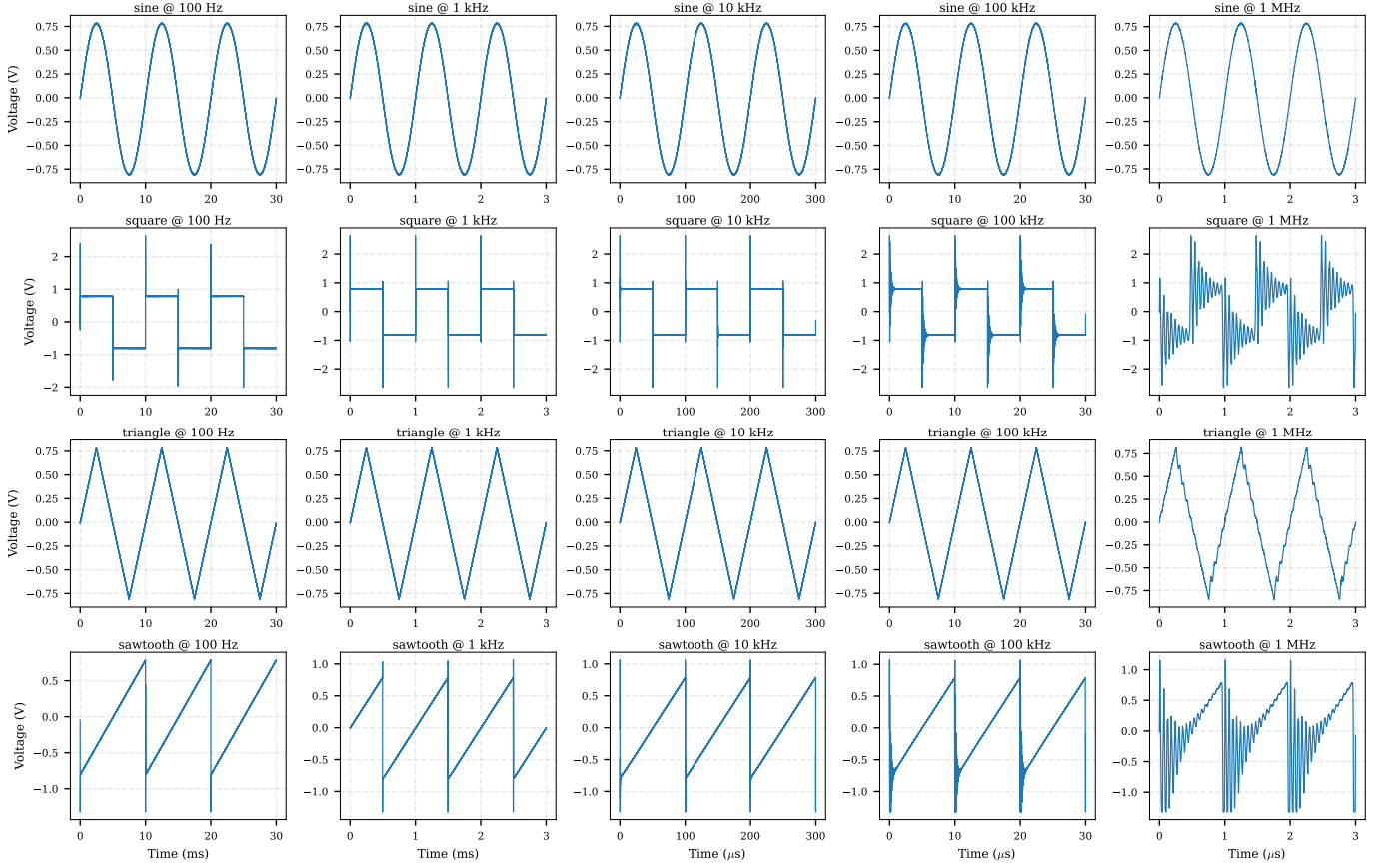


Fig. 10: Oscilloscope-captured preset waveforms: sine, square, triangle, and sawtooth at 100 Hz, 1000 Hz, 10 000 Hz, 100 000 Hz and 1 000 000 Hz. Amplitude set to 0.70 full-scale for all cases.

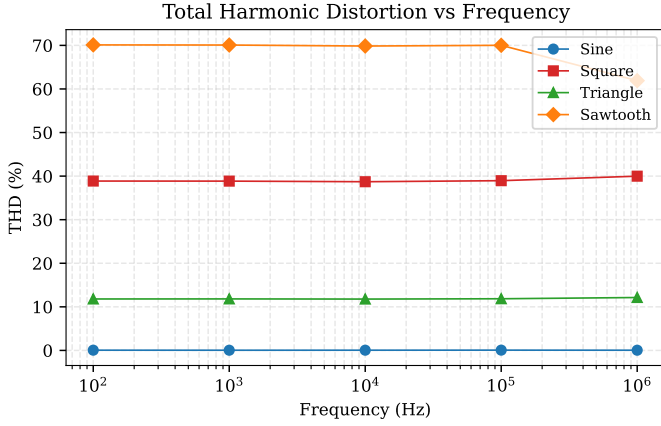


Fig. 11: Measured THD (%) versus frequency for all four preset waveform types. Sine THD remains below 0.07% across the 100 Hz to 1 MHz range.

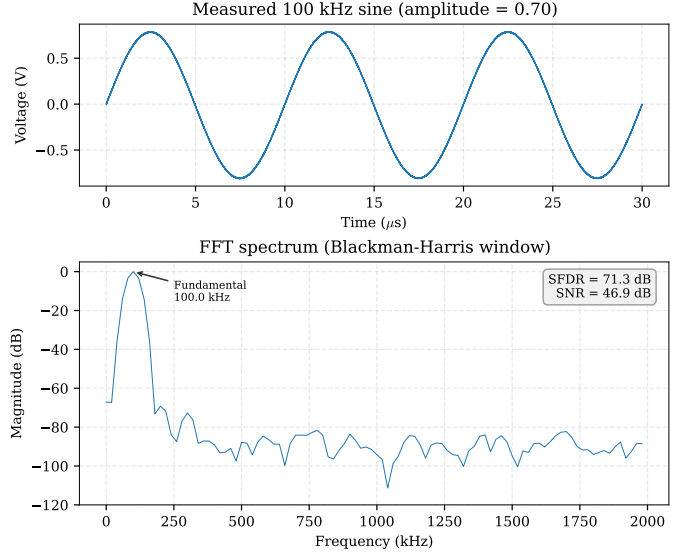


Fig. 12: Measured sine FFT spectrum (100 kHz, amplitude 0.70, Blackman–Harris window). SFDR = 65.5 dBc. The noise floor at approximately -55 dBFS is dominated by oscilloscope quantisation.

E. Dual-Channel Phase Synchronisation

Figure 14 overlays the two DAC channels for programmed phase offsets of 0° , 45° , 90° , and 180° at 1 kHz. Measured

phase errors are summarised in Table V. The worst-case error is 0.0075° , demonstrating sub-millidegree phase accuracy for

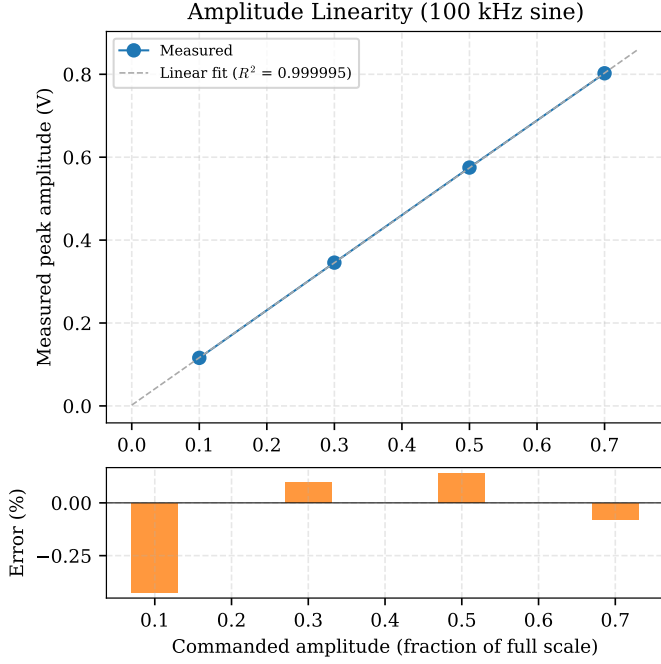


Fig. 13: Amplitude linearity: measured Vpp versus commanded amplitude. $R^2 = 0.999995$. Dashed line shows ideal linear response.

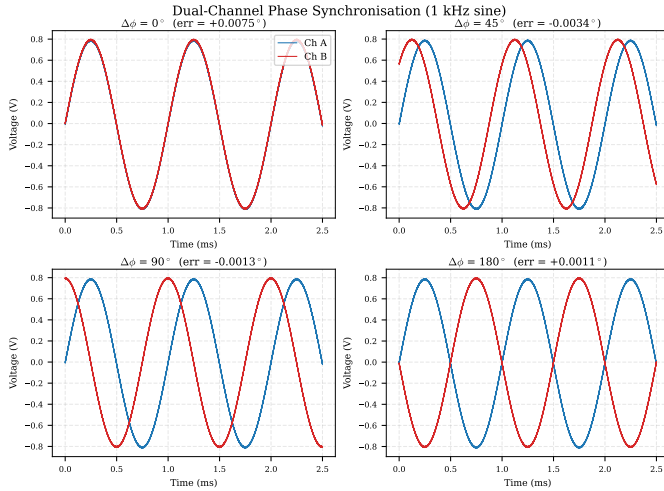


Fig. 14: Dual-channel phase synchronisation at 1 kHz. CH1 (blue) and CH2 (orange) overlaid for four programmed phase offsets.

TABLE V: Dual-Channel Phase Error Measurements

Programmed Offset (°)	Measured Error (°)
0	0.0075
45	-0.0034
90	-0.0013
180	0.0011

the 90° and 180° offsets.

TABLE VI: Performance Summary: Simulation vs. Hardware

Metric	Simulation	Hardware
Sine SFDR	87.6 dBc	65.5 dBc to 69.6 dBc
Sine THD	—	< 0.07%
Freq. accuracy	exact	< 20 ppm
Amplitude linearity R^2	—	0.999995
Phase sync error	exact	< 0.008°
Pipeline latency	7 cycles	7 cycles (56 ns)
DMA fill time (16384 samples)	131 μ s	—
Config apply latency	3–4 cycles	3–4 cycles

TABLE VII: DDS Core FPGA Resource Utilisation (Estimated)

Resource	Usage
BRAM36 (LUT banks: $4 \times 16384 \times 14$)	~32
BRAM36 (Sine ROMs: $2 \times 4096 \times 14$)	~4
DSP48E1 (amplitude multiply)	2
Registers (shadow + active + FSMs)	~800
LUTs (combinational logic)	~600

F. Arbitrary Waveform Fidelity

Figure 15 shows oscilloscope captures of six arbitrary waveform test cases, each driving both DDS channels simultaneously with different but complementary expressions loaded via AXI-DMA. For each case, channel A (blue) carries the primary waveform while channel B (red) provides a related reference—e.g., the AM envelope alongside its modulated carrier, a 3-harmonic square alongside a 7-harmonic one, or a pure sine reference alongside a chirp. All complex waveshapes are faithfully reproduced on both channels concurrently, validating the dual-bank DMA loading path and the independence of the two phase-addressed LUT playback engines.

VIII. PERFORMANCE SUMMARY

Table VI summarises the key performance metrics from both simulation and hardware measurement.

Table VII summarises the estimated FPGA resource utilisation for the DDS core (excluding the Red Pitaya base peripherals). The Zynq-7020 provides 140 BRAM36, 220 DSP48E1, and 106,400 LUTs; the DDS core consumes approximately 26% of BRAM, 1% of DSP, and 0.6% of LUT resources.

IX. CONCLUSION

This report presented the design and verification of a dual-channel DDS IP core with AXI-DMA arbitrary waveform support targeting the Zynq-7020 SoC. The 7-stage pipeline architecture achieves 56 ns latency with sub-micro-hertz frequency resolution. RTL simulations confirm correct operation of all pipeline stages, FSMs, and DMA data paths.

Hardware validation on the Red Pitaya platform demonstrates:

- Sine SFDR of 65.5 dBc to 69.6 dBc and THD below 0.07% across 100 Hz to 1 MHz.
- Amplitude linearity with $R^2 = 0.999995$.
- Dual-channel phase synchronisation within 0.008°.
- Faithful reproduction of complex arbitrary waveforms via the glitch-free dual-bank DMA loading mechanism.

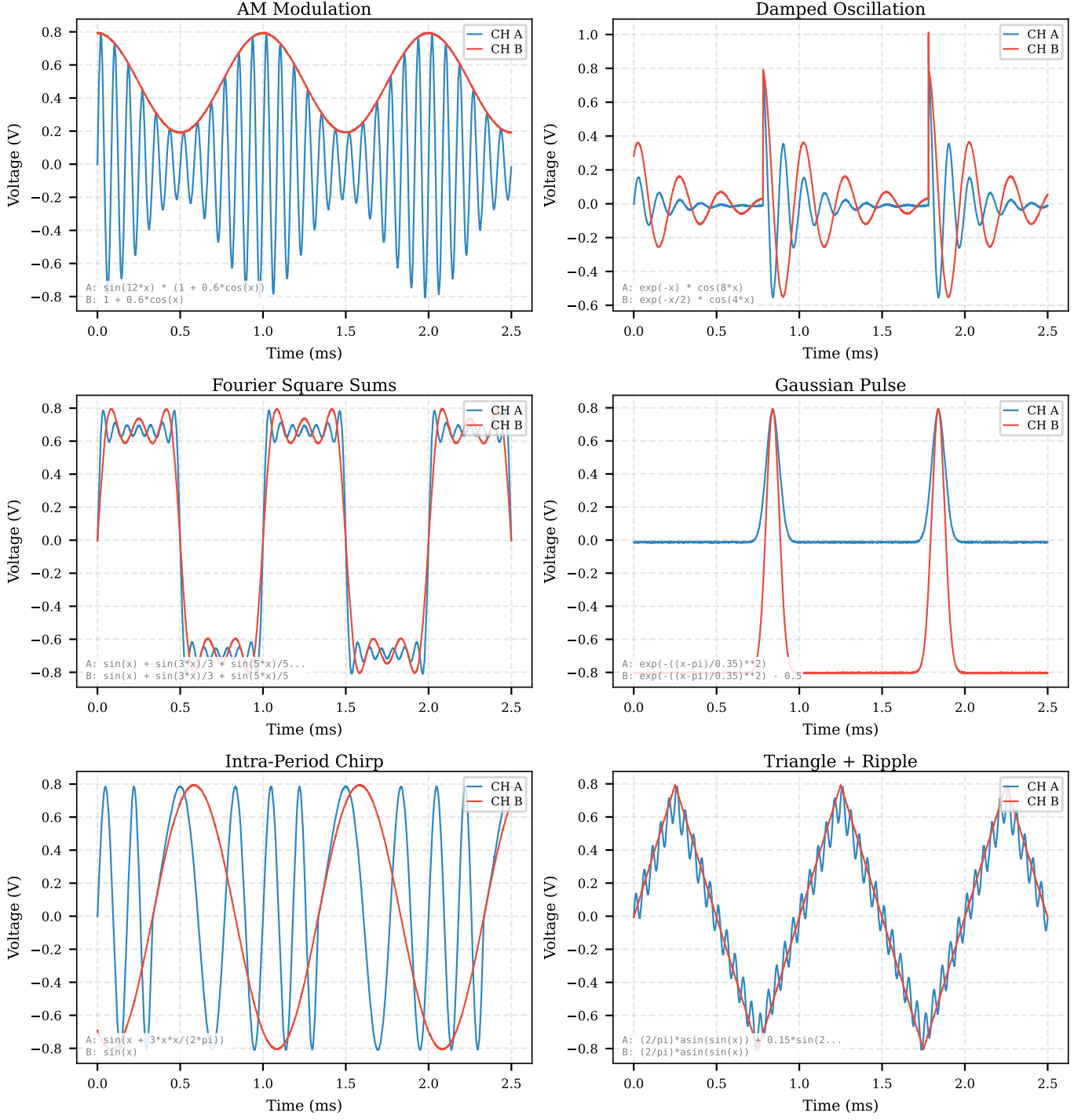


Fig. 15: Dual-channel arbitrary waveforms loaded via AXI-DMA. CHA (blue) and CHB (red) each play a different mathematical expression sampled into independent 16384-entry LUT banks. Per-subplot annotations show the expression programmed on each channel.

The 16 dB SFDR gap between simulation (87.6 dBc) and hardware measurement (65 dBc–70 dBc) is attributed to DAC integral non-linearity, board-level coupling, and the oscilloscope’s own noise floor—consistent with the 14-bit DAC’s theoretical maximum SFDR of approximately 86 dBc.

REFERENCES

- [1] Xilinx, “Zynq-7000 SoC Technical Reference Manual,” UG585 (v1.13), 2022.
- [2] Red Pitaya, “STEMlab 125-14 Hardware Documentation,” <https://redpitaya.readthedocs.io/>, 2024.
- [3] AMD/Xilinx, “AXI DMA v7.1 LogiCORE IP Product Guide,” PG021, 2023.
- [4] E. Murphy and C. Slattery, “All About Direct Digital Synthesis,” *Analog Dialogue*, vol. 38, Aug. 2004.